Non-negativity

Symmetry

Identity

Triangle inequality

Metric Axioms

- **Non-negativity**: $W_p(\alpha, \beta) \geq 0.$
  - Proof: $d(x,y)^p \geq 0$ and $\pi(x,y) \geq 0$, inf of ≥ 0 numbers is ≥ 0
- **Symmetry**: $W_p(\alpha, \beta) = W_p(\beta, \alpha)$
  - Proof: $\pi \in \Pi(\alpha, \beta) \Leftrightarrow \pi^\top(A \times B) := \pi(B \times A) \in \Pi(\beta, \alpha)$. Also, $\int d(x,y)^p \, d\pi = \int d(x,y)^p \, d\pi^\top$ since $d(x,y) = d(y,x)$. Take inf on both sides.
- **Identity**: $W_p(\alpha, \beta) = 0 \iff \alpha = \beta$
  - Proof: One direction ($\Leftarrow$) is easy. For the other one: $\alpha \neq \beta$ implies any valid coupling has nonzero density in at least one pair $(x, y), x \neq y$, so inf is $> 0.$
- **Triangle inequality**: $W_p(\alpha, \beta) \;\leq\; W_p(\alpha, \eta) + W_p(\eta, \beta)$
  - Proof: Harder! Typical proof uses gluing lemma, given $\pi_{\alpha,\eta} \in \Pi(\alpha, \eta)$ and $\pi_{\eta,\beta} \in \Pi(\eta, \beta)$, define a *glued* coupling γ on $\mathscr{P}(\mathscr{X}, \mathscr{X}, \mathscr{X})$ with 1-2 marginal $\pi_{\alpha,\eta}$ and 2-3 marginal $\pi_{\eta,\beta}$. This γ can be marginalized to give a coupling on α, β as well, then use Minkowski and triangle inequality of the ground metric.

So what?